CHAPTER F

Foundations

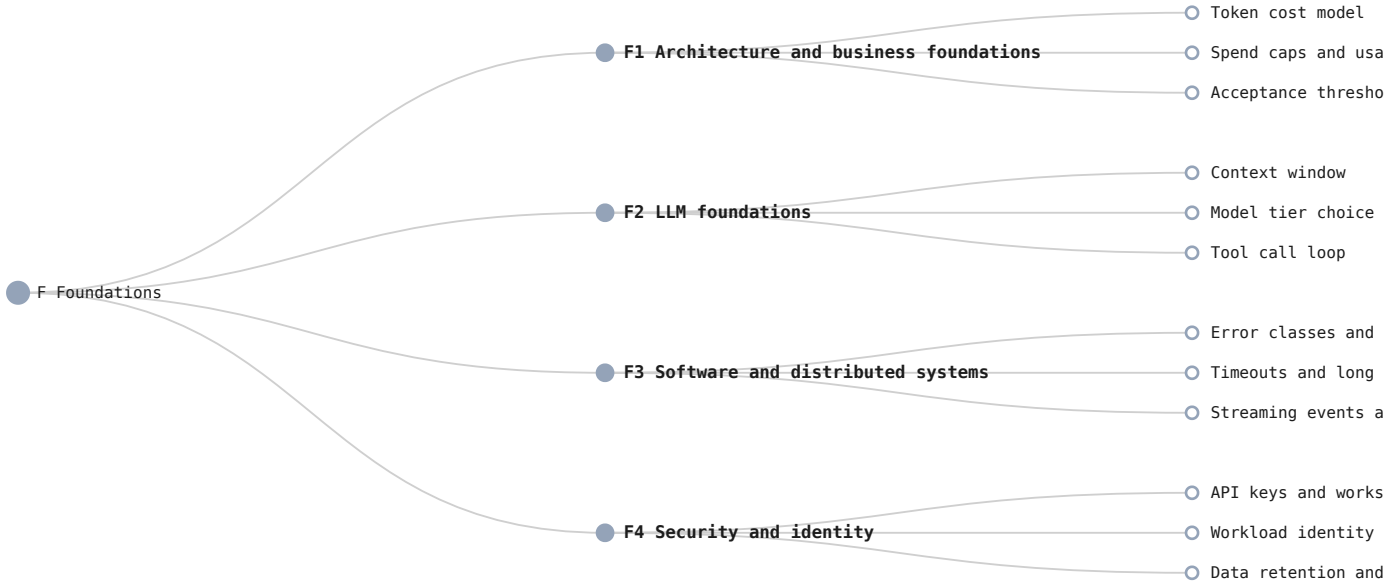
FOUNDATIONS. NOT WEIGHTED.

Four foundations: token cost, context window, errors, keys.

4 sections, 12 components, 12 recall cues.

CONCEPT MAP

Units branch to components. On a small screen, scroll sideways.



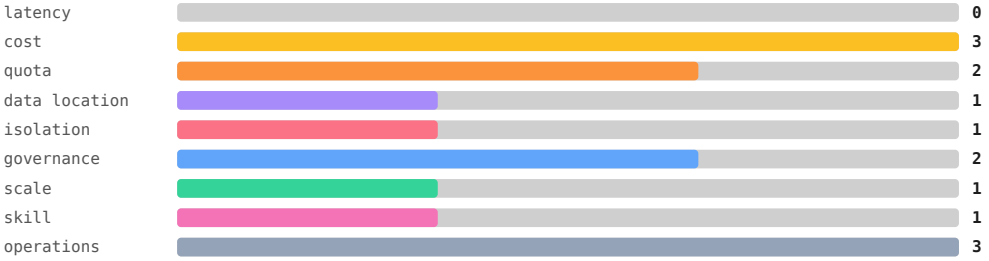
SECTIONS

F1 CHAPTER F, FOUNDATIONS

Architecture and business foundations

Rate per token class, sum the classes, then times volume.

CONSTRAINT FAMILIES



TOKEN COST MODEL

Each token class has its own price, so the cost of a request is the sum of its classes.

Rule Multiply each token class by its own rate, add the classes, then multiply by the request volume.

Trap A flat estimate that counts input tokens only looks simple, but output tokens cost more, so the estimate understates the bill.

Exam tip If a figure has no unit, convert it to tokens per request and requests per day first.

SPEND CAPS AND USAGE TIERS

A spend cap stops usage at a set monthly amount, so it limits the worst-case bill.

Rule Use a workspace spend limit for budget control and a rate limit change for throughput.

Trap A retry loop for every 429 looks right. A spend cap 429 has no retry-after header, so retries fail until access resumes.

Exam tip A 429 with no retry-after header points to a spend cap, not to a rate limit.

ACCEPTANCE THRESHOLDS FOR MODEL OUTPUT

Each quality adjective needs a metric, a threshold, and an owner before the build starts. (Inference)

Rule Attach a metric, a threshold, and an owner to each quality word before the build starts.

Trap Launch first and tune the threshold later sounds agile, but users meet the errors before any bar exists.

Exam tip If the stem says fast or accurate with no number, the answer adds a metric and a threshold.

▼ RECALL CUES (3)

1. **budget for a new workload with a fixed monthly limit**
   Answer workspace spend limit
   Rule Set the budget at the workspace that owns the workload.
   - Spend cap set on the Default Workspace
   - Usage report with no enforcement
2. **estimate the monthly bill from request volume**
   Answer per class token arithmetic
   Rule Multiply each token class by its own rate, then by volume.
   - Input price only
   - One average price for all models
3. **quality target for output that carries money risk**
   Answer metric, threshold, and owner chain
   Rule Write the metric, threshold, and owner before the build.
   - Build first and tune later
   - Quality adjective with no number

▼ NUMBERS AND VERSIONS (15)

- Claude Sonnet 5.5 costs 2 dollars per million input tokens and 10 dollars per million output tokens. source
- Claude Opus 5.5 costs 4 dollars per million input tokens and 20 dollars per million output tokens. source
- Claude Haiku 5.5 starts at 0.10 dollars per million input tokens and 0.50 dollars per million output tokens. source
- Batch API requests cost 50 percent less than the base price. source
- Regional and multi-region endpoints include a 10 percent premium over global endpoints. source
- Setting inference_geo to us on Claude 4.6 and later models applies a 1.1 times pricing multiplier. source
- Prompts over 100,000 tokens cost more on Claude Haiku 5.5. source
- The Start tier monthly spend cap is 500 US dollars. source
- The Build tier monthly spend cap is 1,000 US dollars. source
- The Scale tier monthly spend cap is 200,000 US dollars. source
- A tier spend cap returns HTTP 429 with no retry-after header. source
- A spend limit that you set returns HTTP 400 with an invalid_request_error type. source
- An organization has up to 100 workspaces by default. source
- On Claude Platform on AWS, spend can take about 2 hours to reflect recent usage, and the overshoot is billed. source
- Even with temperature set to 0, identical inputs can produce different outputs across API calls. source

F2 CHAPTER F, FOUNDATIONS

LLM foundations

Fit the window first. Then choose what enters it.

CONSTRAINT FAMILIES



CONTEXT WINDOW

The context window holds every input token and every output token of a turn, so all of them share one budget.

Rule Fit the request inside the window with room for max_tokens, then select what enters the window.

Trap A bigger window looks like the fix for missed details, but context rot means more tokens can lower recall.

Exam tip If the input alone exceeds the window, the API returns a 400 error and no output.

MODEL TIER CHOICE

Each tier trades speed and price against capability, so the choice follows the quality target and the budget. (Inference)

Rule Choose the least costly tier that passes the evaluation for the task.

Trap The largest model looks safest, but it adds cost and latency that the scenario does not require.

Exam tip If the stem names a platform, check the model ID for that platform before you accept an option.

TOOL CALL LOOP

The model requests a tool call, and your code runs the operation and returns the result.

Rule Run every client tool in your code and return each result in a tool_result block.

Trap A design where the model calls the external API directly looks simpler. The model never runs anything on its own, so your app must run the client tool.

Exam tip If stop_reason is tool_use, run the tools before you read the text as the final answer.

▼ RECALL CUES (3)

1. **a long document must be answered in one request**
   Answer window fit check before the design
   Rule Confirm that the input plus max_tokens fits the window, then curate the context.
   - Bigger window as the only fix
   - Input larger than the window
2. **the model must change a record or fetch fresh data**
   Answer client tool loop with tool_result
   Rule Run the tool in app code and return a tool_result block.
   - Model calls the API directly
   - Prompt asks the model to act
3. **high volume with a quality floor and low latency**
   Answer smallest tier that passes the test set
   Rule Pick the least costly tier that passes the evaluation.
   - Largest model by default
   - Model ID copied from another platform

▼ NUMBERS AND VERSIONS (11)

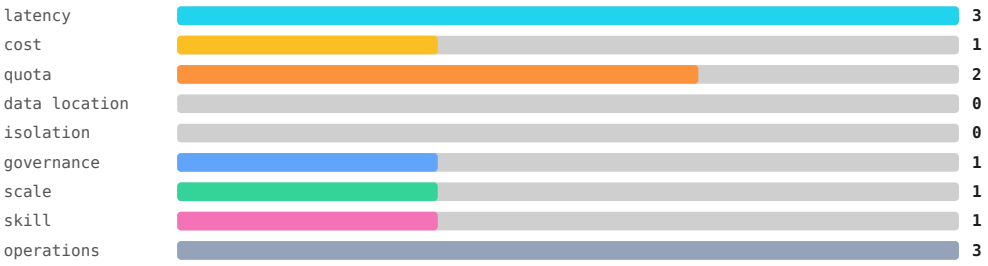
- Claude Opus 5.5 has a 1M-token context window. source
- Models with a 1M-token context window generate up to 128k output tokens in one request. source
- Other models, such as Claude Sonnet 4.5, have a 200k-token context window. source
- An input that alone exceeds the window returns a 400 invalid_request_error. source
- 1M tokens is roughly 555k words on the current tokenizer. source
- One request can include up to 600 images or PDF pages on models with a 1M-token window. source
- Claude Haiku 5.5 is built for high-volume, latency-sensitive tasks such as classification and routing. source
- Claude Haiku 5.5 has a 1M-token context window and up to 128K output tokens. source
- Claude Haiku 5.5 uses a tokenizer that counts about 30 percent more tokens than Claude Haiku 4.5 for the same text. source
- Claude Sonnet 5.5 costs 2 dollars per million input tokens and 10 dollars per million output tokens. source
- Claude Sonnet 5.5 retires not sooner than September 28, 2027. source

F3 CHAPTER F, FOUNDATIONS

Software and distributed systems

Retry 429 and 5xx with backoff. Fix 400 and 401 first.

CONSTRAINT FAMILIES



ERROR CLASSES AND RETRY

Each error type tells the caller whether a retry can help.

Rule Retry 429 and 5xx errors with a limit and backoff. Fix 400 and 401 errors first.

Trap A prompt that says try again looks like a retry policy, but the model cannot resend the request or read a retry-after header.

Exam tip A 429 with retry-after means wait. A 429 with no retry-after header points to a spend cap.

TIMEOUTS AND LONG REQUESTS

A request that runs too long can lose its connection, so long work uses a stream or a batch.

Rule Stream when the user waits, and batch when the caller can poll later.

Trap A longer client timeout looks like the fix for long calls, but idle connections can drop on some networks no matter the timeout.

Exam tip A Messages request that does not stream and may run over 10 minutes needs streaming or a batch.

STREAMING EVENTS AND ERRORS

A stream sends typed events after a 200 status, so an error can arrive inside the stream.

Rule Handle error events inside the stream loop. Do not rely on the HTTP status check alone.

Trap Only the HTTP status code looks sufficient, but a stream can fail after the 200 status.

Exam tip An overloaded_error event inside a stream corresponds to HTTP 529 in a call that does not stream.

▼ RECALL CUES (3)

1. **a call fails with a server error and the request is valid**
   Answer bounded retry with backoff in caller code
   Rule Retry transient errors with a limit. Fix invalid requests first.
   - Prompt asks the model to retry
   - Retry loop with no limit
2. **a long job that no user waits for**
   Answer batch with polling
   Rule Create a batch and poll processing_status until it is ended.
   - Longer client timeout
   - Thousands of synchronous calls in a loop
3. **partial output appears, then the call fails**
   Answer stream event handling
   Rule Handle error events inside the stream loop.
   - Check only the HTTP status
   - Restart the job without the request ID

▼ NUMBERS AND VERSIONS (14)

- The official SDK retries connection errors, rate limits, and 5xx errors twice by default. source
- A 500 api_error means an unexpected internal error, and the docs say to retry with exponential backoff. source
- A 529 overloaded_error means the API is temporarily overloaded. source
- The Messages API accepts a request up to 32 MB. source
- A 413 request_too_large error means the request exceeds the size limit. source
- The SDK validates that Messages requests that do not stream are not expected to exceed a 10-minute timeout. source
- Most batches complete within 1 hour. source
- Batches expire if processing does not complete within 24 hours. source
- Batch results are available for 29 days after creation. source
- The Messages API accepts a request up to 32 MB. source
- The Batch API accepts a request up to 256 MB. source
- An error can occur after the API returns a 200 response during a server-sent event stream. source
- An overloaded_error in a stream would normally correspond to HTTP 529. source
- Resume of an interrupted stream is documented for Claude 4.5 models and earlier. source

F4 CHAPTER F, FOUNDATIONS

Security and identity

Scoped, time-limited, identity-backed key. Never in the prompt.

CONSTRAINT FAMILIES



API KEYS AND WORKSPACE SCOPE

An API key is a bearer secret, so its scope and expiration limit the damage from a leak.

Rule Use a scoped, time-limited, identity-backed key, and store it in a secrets manager.

Trap A key in a prompt or a config file looks convenient, but the model reads the prompt and the file exposes the secret.

Exam tip A 401 authentication_error can mean a revoked, malformed, or expired key.

WORKLOAD IDENTITY FEDERATION

A workload presents a signed token from its own identity provider and receives a short-lived Anthropic token in return.

Rule Use federation when the workload has a trusted identity, and use a personal key only for development.

Trap A long-lived key in CI secrets looks simpler, but it is a static secret that stays valid until someone rotates or disables it.

Exam tip A jti_reused failure means the JWT was presented before, so mint a new JWT for the exchange.

DATA RETENTION AND ENCRYPTION

A data control names who stores data, for how long, and which key protects it.

Rule Enable ZDR or CMEK only through the account team, after you confirm the eligible features.

Trap Platform logs look like a data control, but logs do not enable zero data retention.

Exam tip CMEK enablement is permanent, so the answer confirms the key plan before the change.

▼ RECALL CUES (3)

1. **a CI job needs API access without a stored secret**
   Answer federation with short-lived token
   Rule Federate the workload identity when the platform issues one.
   - Long-lived key in CI secrets
   - Prompt rule that says keep the key private
2. **a team shares one script that uses a personal key**
   Answer service account key
   Rule Create a service account key for shared workloads.
   - Personal key shared by a team
   - Legacy workspace key as the default
3. **regulated data must not be stored at rest**
   Answer ZDR through the account team
   Rule Request ZDR after you confirm that the feature is eligible.
   - Console playground for sensitive data
   - Activity logs as a substitute for ZDR

▼ NUMBERS AND VERSIONS (13)

- Key expiration is a preset of 3 hours, 1 day, 7 days, or 30 days, a custom duration, or Never. source
- An expired key returns 401 authentication_error and cannot be reactivated. source
- Anthropic emails the key creator 7 days before expiration for keys with a lifetime of at least 14 days. source
- Admin API keys start with sk-ant-admin01-. source
- Short-term bearer tokens on Amazon Bedrock have a 12-hour maximum. source
- The default token lifetime in a federation rule is 3,600 seconds. source
- The minted token lifetime is never less than 60 seconds. source
- The SDK starts an advisory refresh at expiry minus 120 seconds. source
- The SDK starts a mandatory refresh at expiry minus 30 seconds. source
- The federation setup wizard listens for a token exchange for 15 minutes. source
- A federation rule accepts token_lifetime_seconds from 60 to 86400, with a default of 3600. source
- Covered Models require 30-day data retention. source
- Anthropic recommends logging activity on at least a 30-day rolling basis. source